Controlled Vocabularies for a Knowledge Graph on Open Educational Resources

Steiner, Petra C.
petra.steiner@tu-darmstadt.de
Technical University of Darmstadt

Geiger, Jonathan
jonathan.geiger@adwmainz.de
Academy of Sciences and Literature Mainz

Lange, Frank
f.lange@itc.rwth-aachen.de
RWTH Aachen University

Desouki, Abdelmoneim Amer
abdelmoneim.desouki@tu-darmstadt.de
Technical University of Darmstadt

The scientific infrastructure project DALIA ("Data Literacy Alliance") aims to index, link, visualize, and make accessible existing open educational resources (OER) on data literacy. It builds and maintains a platform that encompasses materials for the National Research Data Infrastructure (NFDI), federal Research Data Management (RDM) initiatives, data competence centers, and beyond. The different needs of career and competence levels and disciplines must be considered and integrated. The scopes range from professor to student, and the disciplines cover a huge diversity with fields such as history, life sciences, or engineering.

For the linking and integration of the learning resources, DALIA is developing a knowledge graph that establishes the adherence to the FAIR Principles (cf. Garcia et al. 2020). To ensure the interoperability of teaching and learning materials for this knowledge graph, a schematization of their metadata is required: The DALIA Interchange Format (DIF) provides a framework for making the metadata comparable and smoothly integrable into the DALIA platform. It describes and explains the metadata fields for the online publication of educational resources (Geiger et al. 2025). The selection of descriptive elements is based on the Dublin Core Application Profile (DCTAP) (Coyle 2022). Cardinality and value constraints of attributes are specified. The DIF consists of elements compiled from other namespaces, especially for attributes and controlled vocabularies. Most attributes of DIF 1.3 (Geiger et al. 2024), such as *authors*, *title*, *description*, *language*, *keywords*, and *discipline* refer to the standards *dcterms* (DCMI Usage Board 2020), *schema* (schema.org project 2024), or *fabio* (Shotton / Peroni 2019). In cases where standardized definitions are lacking, custom terms are defined in the underlying MoDALIA ontology (BMBF DALIA project 2024). For the DIF, different mappings were developed, e.g. for schema.org and the AMB (Pohl et al. 2024).

The development of picklists, however, presents a more complex challenge: While some picklists can be compiled from existing controlled vocabularies, such as licenses from the SPDX License List (SPDX 2024), defined by the class *spdx:ListedLicense* from the SPDX terms (SPDX 2022), others deal with issues of vagueness. For example, the assignment of values for LearningResourceTypes raises questions about the distinction between tutorials and lectures. Furthermore, the diversity of disciplines and the various forms of materials necessitated the development of context-specific picklists. In this community-driven process, each approach and field are appreciated and considered. If definitions of picklist items are missing, they are defined in the MoDALIA ontology.

For technical requirements, a description schema has been developed, comprising the attribute names or picklist items, definitions, sources, IRIs of the terms, and relations to other terms.

This poster focuses on the description schema and the compilation of picklists for the DALIA Interchange Format, with the aim of facilitating the description and classification of teaching and learning material. It aims to deepen communication with the international professional community by inviting all interested parties to feedback and to get in touch.

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